Saeed Mohseni seh deh

saeedmohseniph@gmail.com | 540-558-5622 | Personal Website

LinkedIn | Google Scholar | GitHub

Summary

PhD student in Electrical Engineering at Virginia Tech researching causal learning, generative modeling, and deep learning. Highly interested in AI/ML research and engineering roles where rigorous modeling meets large-scale, real-world systems.

Education

Virginia Tech Dec 2023 - Dec 2027
PhD in Electrical Engineering, Advisor: Dr. Walid Saad CGPA: 4.0/4.0
Research: Causal Learning and Interpretable AI with Mathematical Performance Guarantees

Sharif University of Technology Sep 2020 - Sep 2022
MSc in Electrical Engineering, Advisor: Dr. Massoud Babaie-Zadeh CGPA: 3.7/4.0
Thesis: Dictionary Learning for Sparse Representation-based Classification

Amirkabir University of Technology Sep 2016 - Sep 2020
BSc in Electrical Engineering, Advisor: Dr. Hamidreza Amindavar CGPA: 4.0/4.0
Thesis: Sketching for Sequential Change-point Detection

Professional Experience

Graduate Research Assistant | Virginia Tech **Dec 2023 – Present**

- Developed a diffusion-based method for inverse problems using piecewise guidance and multi-GPU parallelization in PyTorch, achieving $\geq$23% faster recovery on ImageNet with comparable PSNR/SSIM. (**S. Mohseni-Sehdeh et al.**, *TMLR, 2026*)[1]
- Designed a novel causal discovery algorithm leveraging our newly introduced induced covariance, achieving 35% higher precision and 41.5% higher recall than existing methods.[2]
- Engineered a modular digital twin-based control system for signal tracking tasks, achieving 49% lower MSE than classical control methods and 16% lower MSE when using causal learning generated training data versus statistical learning data in low data self training scenarios. (**S. Mohseni-Sehdeh et al.**, *CISS, 2025*) [3]

Graduate Teaching Assistant, *Senior Design Project* | Virginia Tech **Jan 2024 – Dec 2024**

- Mentored and evaluated 8 student teams (32 students) on industry-oriented engineering projects and helped organize a university-wide project expo.

Graduate Research Assistant | Sharif University of Technology **Aug 2021 – Jun 2023**

- Developed 2 novel algorithms in sparse signal processing and overcomplete dictionary learning for signal classification, improving accuracy over baseline methods.
- Developed a resource-efficient dictionary-based classification algorithm using undercomplete dictionaries, achieving 10× faster inference and 100× faster training than competing methods while maintaining or improving accuracy. (**S. Mohseni-Sehdeh et al.**, *Signal Processing, 2023*) [4]

Publications

1. S. Mohseni-Sehdeh, W. Saad, K. Sakaguchi, and T. Yu,"Diffusion Models for Solving Inverse Problems via Posterior Sampling with Piecewise Guidance," *Transactions on Machine Learning Research (TMLR)*, Apr. 2026 [OpenReview] [Code]
2. S. Mohseni-Sehdeh and W. Saad, "Induced Covariance for Causal Discovery in Linear Sparse Structures", *IEEE Signal Processing Letters*, under review. [arXiv]
3. S. Mohseni-Sehdeh, W. Saad, K. Sakaguchi, and T. Yu, "Causal Digital Twins for Physical System Control,"in *Proc. of 59th Annual Conference on Information Sciences and Systems (CISS)*, pp. 1–6, Mar. 2025. [DOI]
4. S. Mohseni-Sehdeh and M. Babaie-Zadeh, "A fast dictionary-learning-based classification scheme using undercomplete dictionaries," *Signal Processing*, vol. 212, p. 109124, Nov. 2023. [DOI] [Code]

Selected Projects

TinyStories LLM, Building a Small Language Model from Scratch | **Github repository**

- Built and trained a 4.8M parameter **GPT-style transformer** LLM from scratch using PyTorch on the TinyStories dataset reaching final validation loss of 1.4.
- Implemented multi-head attention, positional embeddings, and layer normalization manually.
- Gained deep understanding of transformer internals and training dynamics.

Joint Object and Depth Detection on NYU Depth Dataset V2

- Developed a pipeline for simultaneous object detection and depth estimation on the NYU Depth Dataset V2, using DenseNet-169 (14M parameters, ImageNet pretrained) as a feature extractor.
- Applied **YOLO** for object detection and a CNN-based model for per-object depth estimation, achieving a validation MSE of 1.21.
- Conducted experiments evaluating accuracy and multi-task learning performance, demonstrating robust integration of computer vision techniques on high-dimensional image data.

**Additional projects and reports are available here.*

Technical & Analytical Skills

Programming Languages: Python, MATLAB, C

AI / ML Frameworks & Tools: PyTorch, TensorFlow, NumPy, Pandas, Scikit-learn.

Quantitative & Analytical Tools: Statistical Modeling, Data Analysis, Numerical Optimization, Hypothesis Testing, Time-Series Modeling, Bayesian Inference.

Other Tools: Git, LaTeX, MATLAB Simulink.

Relevant Coursework: Deep Learning (CNN, RNN, LSTM), Data Analytics, Linear Algebra, Numerical Optimization Methods, Numerical Analysis, Probability Theory, Statistical Inference, Random Processes, Digital Signal Processing.

Honors & Certificates

- **Fully Funded Ph.D. Student**, Virginia Tech, USA (Present).
- **Fully Funded M.Sc. in Electrical Engineering**, admitted as an *Exceptional Talent Student* to the top 1 university in Iran.
- **Ranked 2nd** in B.Sc. Electrical Engineering class, Amirkabir University of Technology, Iran.
- **Top 0.6% nationally** in the Mathematics–Physics track of the Iranian National University Entrance Exam.
- **Reinforcement Learning Specialization**, Coursera (University of Alberta, Jun 2022). [Link]
- **Generative Adversarial Networks (GANs) Specialization**, Coursera (DeepLearning.AI, Oct 2021). [Link]
- **Deep Learning Specialization**, Coursera (DeepLearning.AI, Aug 2021). [Link]